Experiment 11:

Linearity and Time Invariance

```
clc;

clear;

close all;

% Time Vector t = -1:0.001:1;

% Input Signals x1 = sin(2pi5t); x2 = cos(2pi5t);

% Parameters a = 2; b = 3; t0 = 0.2;

% System Definition T = @(x) 2*x;

% Linearity Test lhs = T(ax1 + bx2); rhs = aT(x1) + bT(x2);

linError = max(abs(lhs - rhs));

% Time-Invariance Test xDelay = sin(2pi5*(t - t0));

lhsTI = T(xDelay); rhsTI = T(x1);

rhsTIshift = interp1(t, rhsTI, t - t0, 'linear', 0);

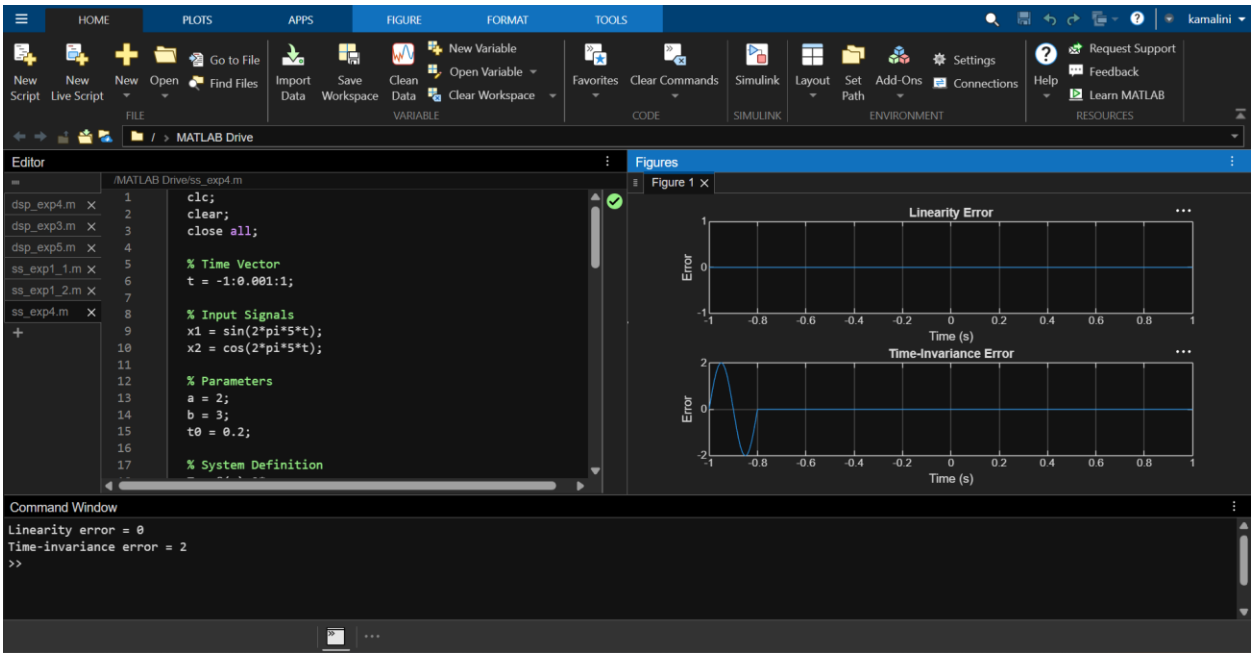
tiError = max(abs(lhsTI - rhsTIshift));

% Plot Results figure;

subplot(2,1,1); plot(t, lhs - rhs); grid on; title('Linearity Error'); xlabel('Time (s)'); ylabel('Error');

subplot(2,1,2); plot(t, lhsTI - rhsTIshift); grid on; title('Time-Invariance Error'); xlabel('Time
(s)'); ylabel('Error');

% Display Results fprintf('Linearity error = %g\n', linError); fprintf('Time-invariance error
= %g\n', tiError);
```


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kamalini
New Script
New Live Script
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FILE
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Clean Data
New Variable
Open Variable
Clear Workspace
VARIABLE
Favorites
Clear Commands
CODE
Simulink
SIMULINK
Layout
Set Path
Add-Ons
Settings
Connections
ENVIRONMENT
Help
Request Support
Feedback
Learn MATLAB
RESOURCES
/ › MATLAB Drive
Editor
/MATLAB Drive/ss_exp4.m
dsp_exp4.m
dsp_exp3.m
dsp_exp5.m
ss_exp1_1.m
ss_exp1_2.m
ss_exp4.m
clc;
clear;
close all;
% Time Vector
t = -1:0.001:1;
% Input Signals
x1 = sin(2*pi*5*t);
x2 = cos(2*pi*5*t);
% Parameters
a = 2;
b = 3;
t0 = 0.2;
% System Definition
Figures
Figure 1
Linearity Error
Error
Time (s)
Time-Invariance Error
Error
Time (s)
Command Window
Linearity error = 0
Time-invariance error = 2
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